

Sam Cook

sgcook97@gmail.com | (512) 647-9578 | <https://samcook.vercel.app>

Austin, TX 78727

EDUCATION

Texas State University

Bachelor of Science in Computer Science

Dec 2023

San Marcos, TX

Relevant Coursework: Machine Learning, Parallel Programming, Database Systems, Recommender Systems, Cyber Security, Data Structures and Algorithms, Object Oriented Programming

EMPLOYMENT EXPERIENCE

Abrigo

May 2025-Present

Technical Support Analyst

Austin, TX

- Troubleshoot and resolve technical issues for on-premises and hosted financial crimes software, including database errors, server performance, system configurations, and connectivity.
- Investigate database issues, validate data accuracy, and implement solutions to resolve software and system-level problems, while collaborating with internal engineers and cross-functional teams when needed.
- Assist customers with system configurations, server-related troubleshooting, and software functionality while documenting solutions and ensuring reliable system performance and regulatory compliance.

Heartflow

Oct 2024 - May 2025

Case Analyst

Austin, TX

- Created precise 3D models of a patient's heart from CT scans using HeartFlow's software to assist in diagnosing coronary artery disease.
- Verified model accuracy and maintained quality through detailed inspections and process adherence.
- Worked within a production team to meet deadlines and contributed to product improvements through testing and feedback.

TECHNICAL SKILLS

Databases: MongoDB, PostgreSQL, MySQL

Cloud Services: AWS, Vercel, Render

Version Control: Git, Github

Languages: Python, C++, TypeScript, Rust, C#, JavaScript, Java, HTML/CSS.

Frontend Technologies: React, Next.js

Backend Frameworks: Express.js, Django, Flask, Axum

ML/Data Science: TensorFlow, PyTorch, Scikit-Learn, Numpy

PROJECTS

Homelab

- Built and maintain a Proxmox-based virtualization environment, managing Linux containers, networking, storage, and system resources.
- Deployed and maintain self-hosted applications and game servers using Linux, SSH, systemd, ZFS, and network configuration.
- Troubleshoot connectivity, storage, performance, and application issues using system logs, monitoring tools, and command-line utilities.

Blockparty

Frontend (<https://github.com/sgcook97/vball-meetup-client>) API (<https://github.com/sgcook97/vball-meetup-api>)

- Developed a MERN web-app to allow volleyball players to connect and play with people in their area.
- Integrated real-time messaging with Socket.IO for users to communicate with others to plan events, capable of supporting 150+ users at once.
- Engineered an Express.js API that utilizes MongoDB for scalable storage, allowing users to create posts and store old conversations with other users.
- Embedded a weather tracker using Open-Meteo and Nominatim APIs that displays a 24-hour forecast to let users know if conditions are good enough to play volleyball.

SamnRoll

- Developed a full stack anime recommendation web app, incorporating a Flask API and a Next.js frontend.
- Backend (<https://github.com/sgcook97/anime-rec-api>):
 - Engineered a Flask API deployed on AWS EC2, generating personalized anime recommendations utilizing content-based and collaborative filtering models trained using more than 3 million ratings.
 - Integrated PostgreSQL database on AWS RDS, ensuring secure storage data.
 - Established connectivity to the MyAnimeList API to retrieve details on over 100,000 different anime.
- Frontend (<https://github.com/sgcook97/anime-rec-client>):
 - Constructed a Next.js application utilizing TypeScript.
 - Enabled users to search for and rate titles for improved recommendation accuracy.
 - Integrated the backend using responses received from the Flask API to display titles and recommendations, and store user data such as login details and ratings.